

ECON 2:

Principles of Macroeconomics

Lecture 10: Short-Run Macroeconomics



Additional Types of Spending

Spending by businesses

Planned Investment Spending (I^P)

Two Major Categories:

1.

2.

What is “unplanned investment”?





Determinants of Planned Investment Spending (I^P):

- 1.
- 2.
- 3.

Assumption: Output today (Y) doesn't directly affect I^P

As Y changes, I^P is constant!



Spending by the Government (G)

What is included?

- Military
- Fire and Police
- Public Education
- Parks and Rec
- Highway infrastructure

Industries that produce necessary goods/service, but have trouble making a profit

Does not include *Transfer Payments* such as:

- Social Security/Welfare
- Financial Aid
- Subsidies

Assumption: Output today (Y) doesn't directly affect G

As Y changes, G is constant!



Additional Types of Spending

Spending by the Foreigners

Net Exports (NX) = Exports (X) – Imports (M)

Assumption:

Output (Y) in the short-run does not directly affect Net Exports (NX)





Aggregate Expenditures

Total Spending in the Economy = Aggregate Expenditures = AE

AE = Consumption + Planned Investment + Gov't Expenditures + Net Exports

AE = C + I^P + G + NX (Note: I^P = I when change in inventories = 0)

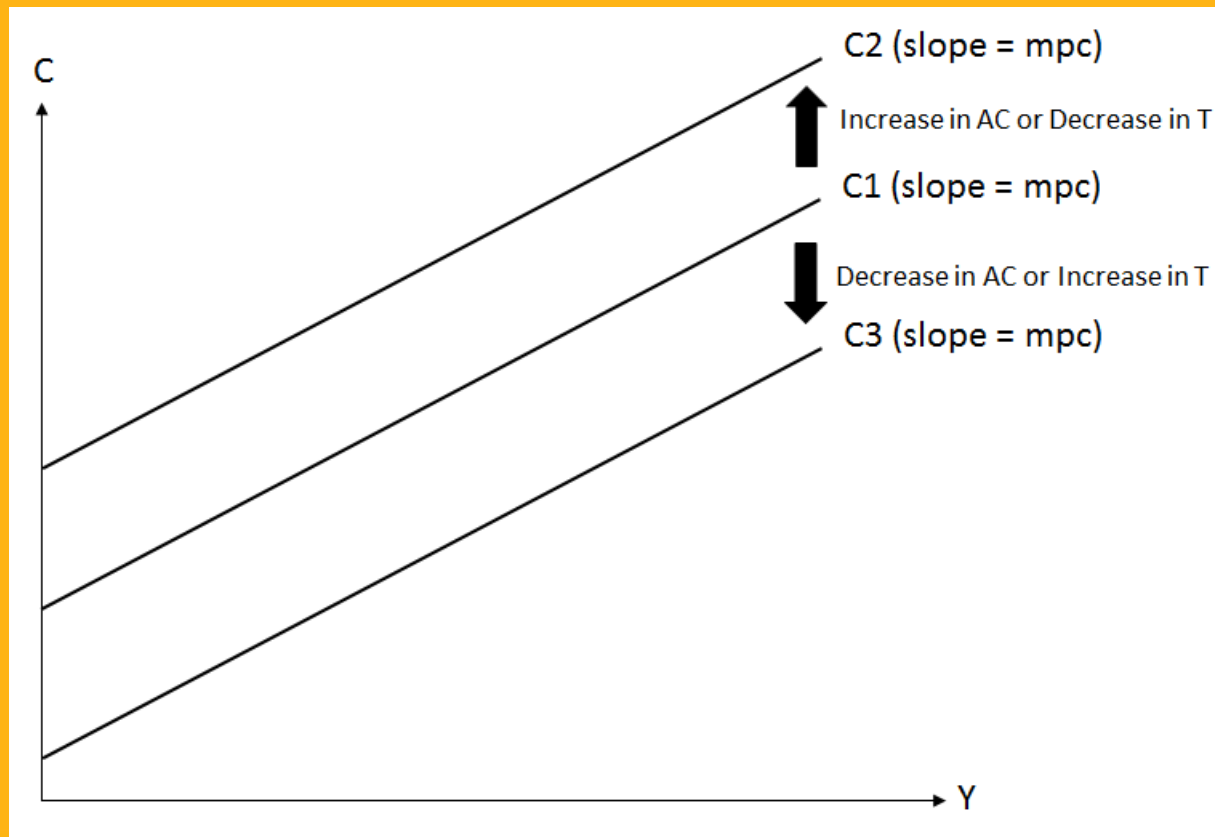
Show how spending (AE) changes when Y changes and find where Y = AE

AE = AC + mpc (Y - T) + I^P + G + NX

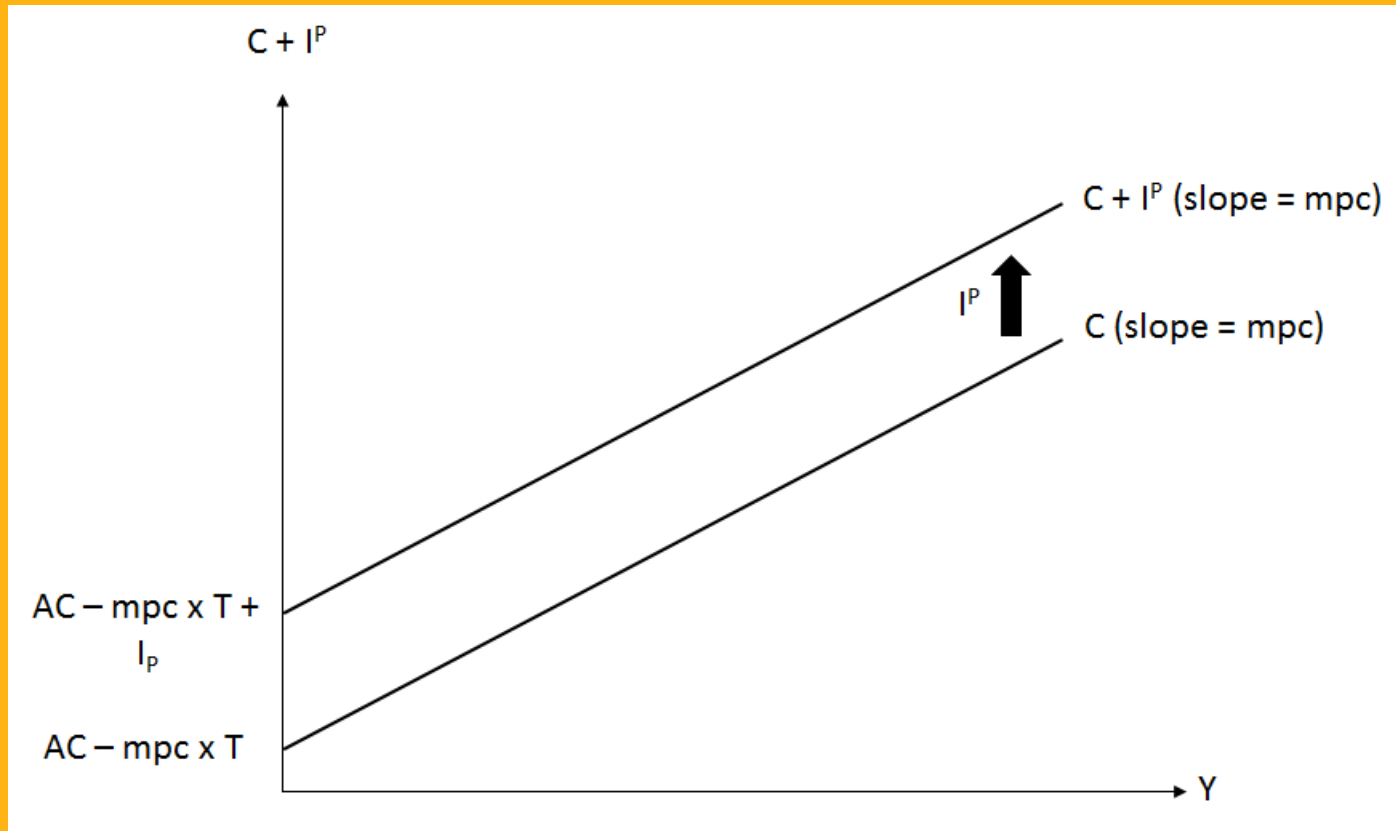


Building the Aggregate Expenditure Curve

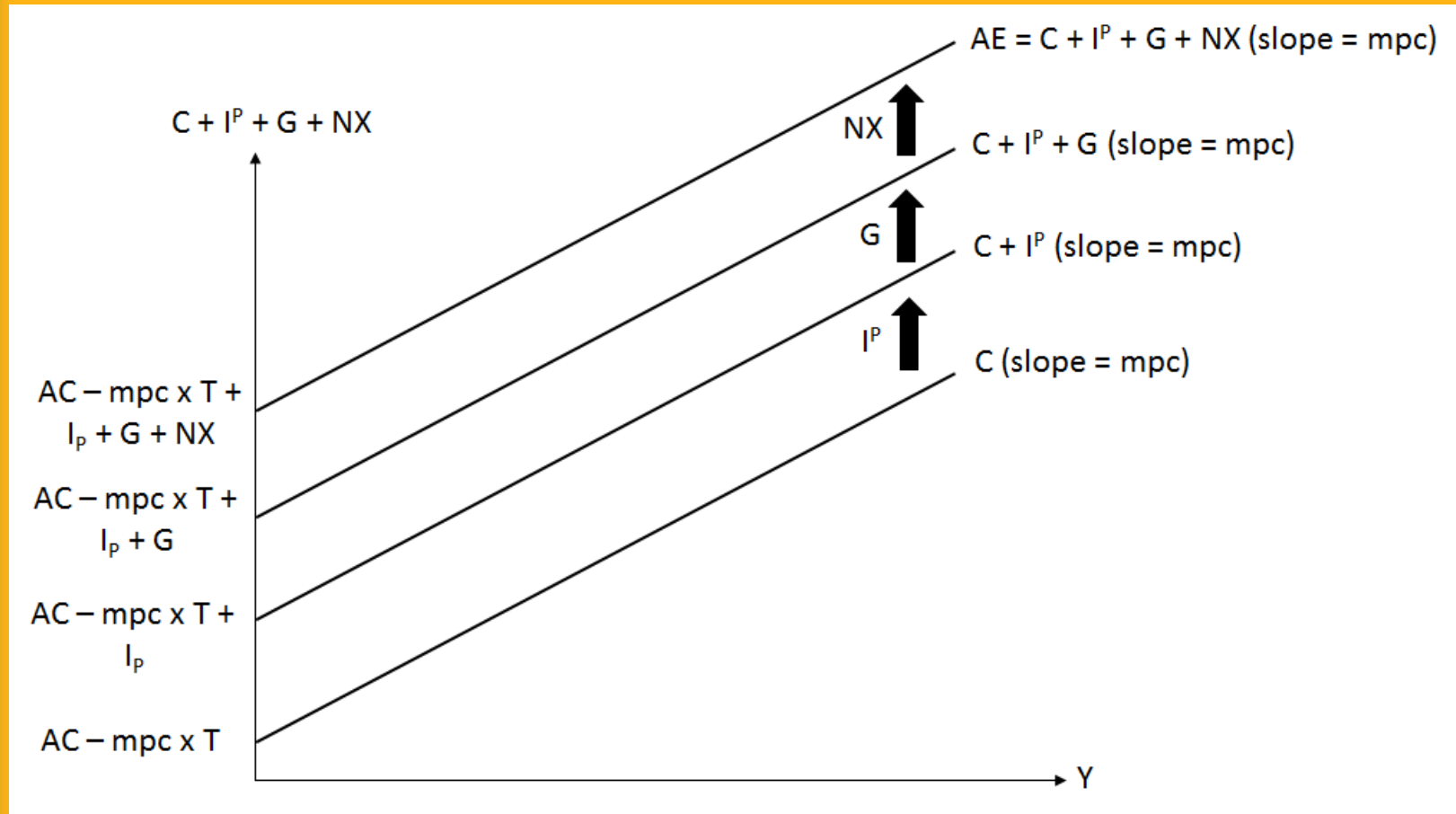
Start with Consumption



Building the Aggregate Expenditure Curve



Building the Aggregate Expenditure Curve



Where is equilibrium?



- Scenario 1: $Y = 12000$, but $AE = 11000$
 - Unsold production?
- Scenario 2: $Y = 8000$, but $AE = 9000$
 - Meeting consumer demands without production?
- Scenario 3: ?



Graphing Equilibrium

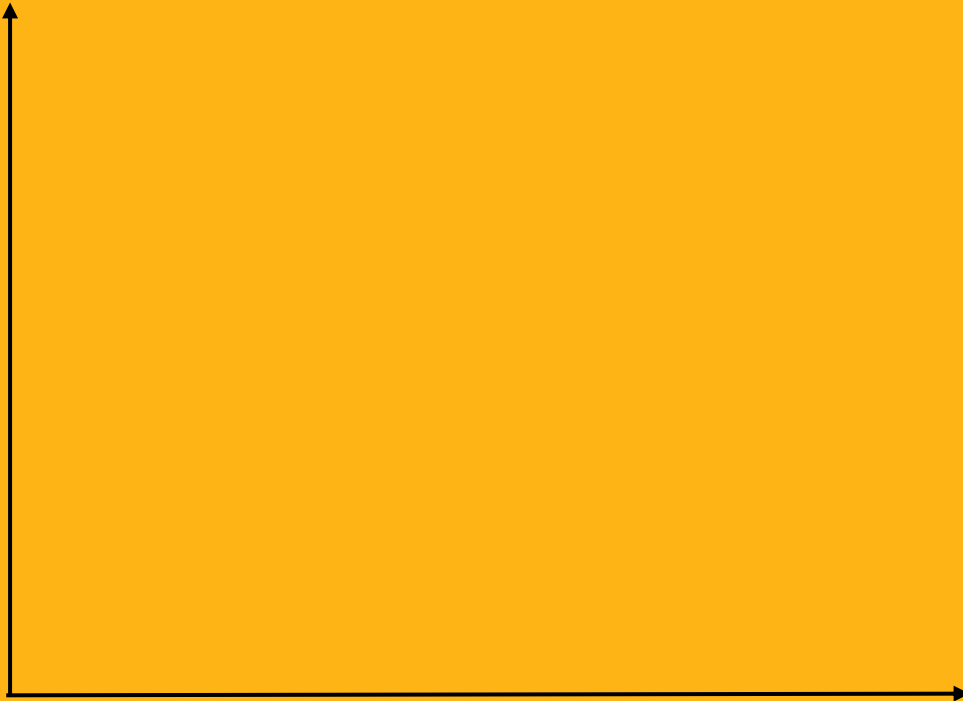


Finding Equilibrium

- Economy:
- $I^P = 10$; $G = 10$; $NX = 5$

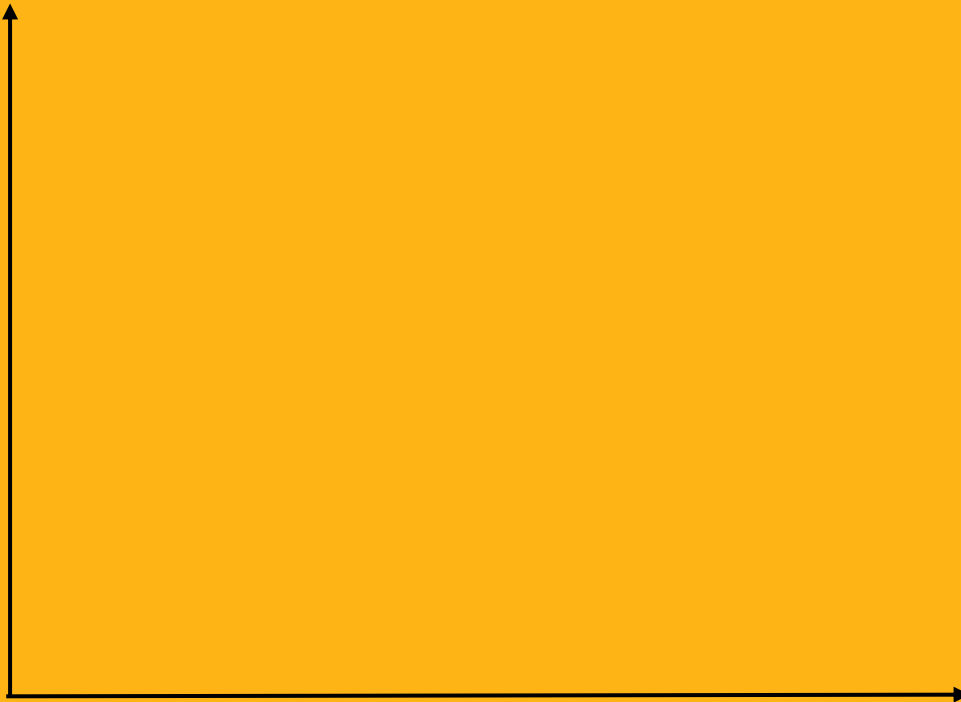
Real GDP (Y)	Consumption (C)	Planned Investment (I^P)	Government Expenditures (G)	Net Exports (NX)	Aggregate Expenditures (AE)	Change in Inventory
200	200	10				
300	275					
400	350			5		
500	425					
600	500		10			

Graphing Equilibrium



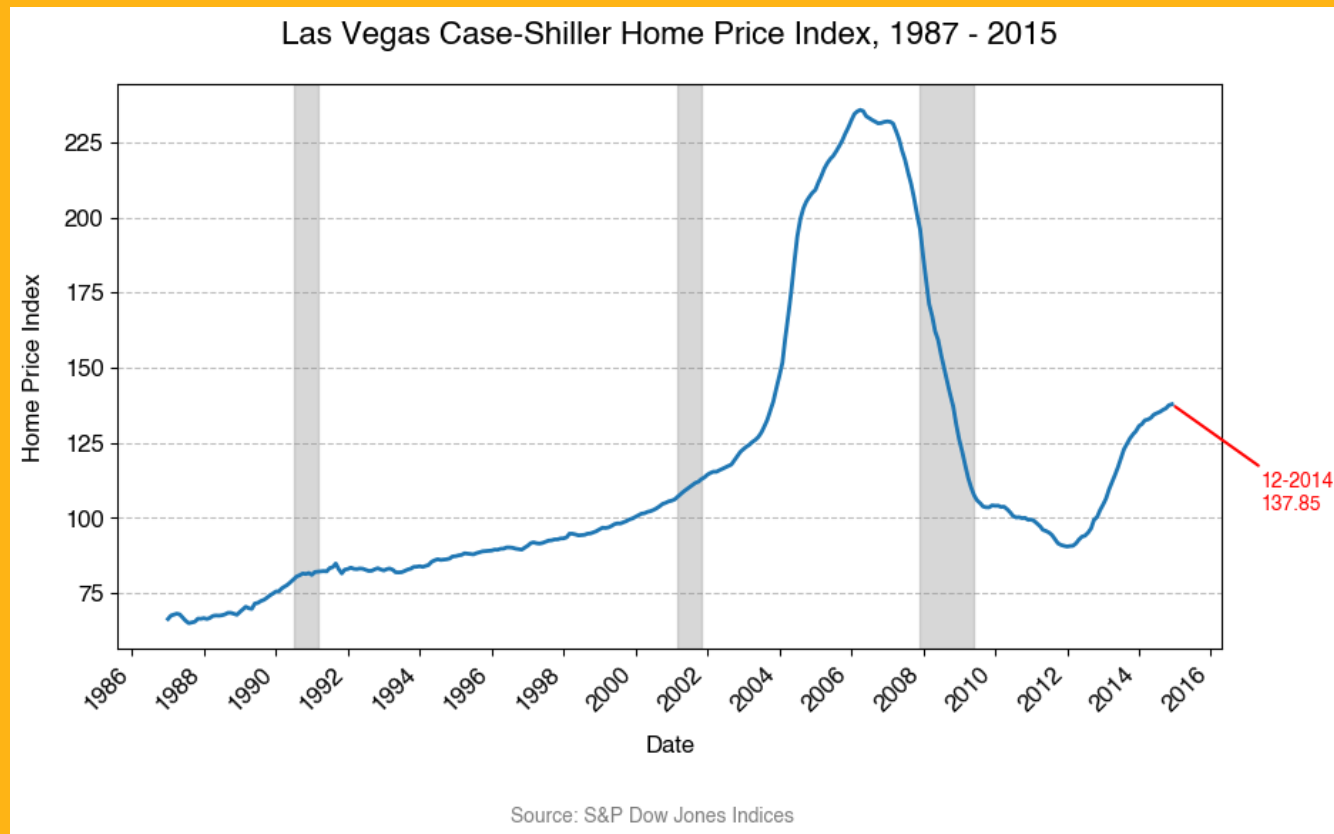
Changes in Equilibrium

- Start at $Y^* = 10,000$ (full-employment output)



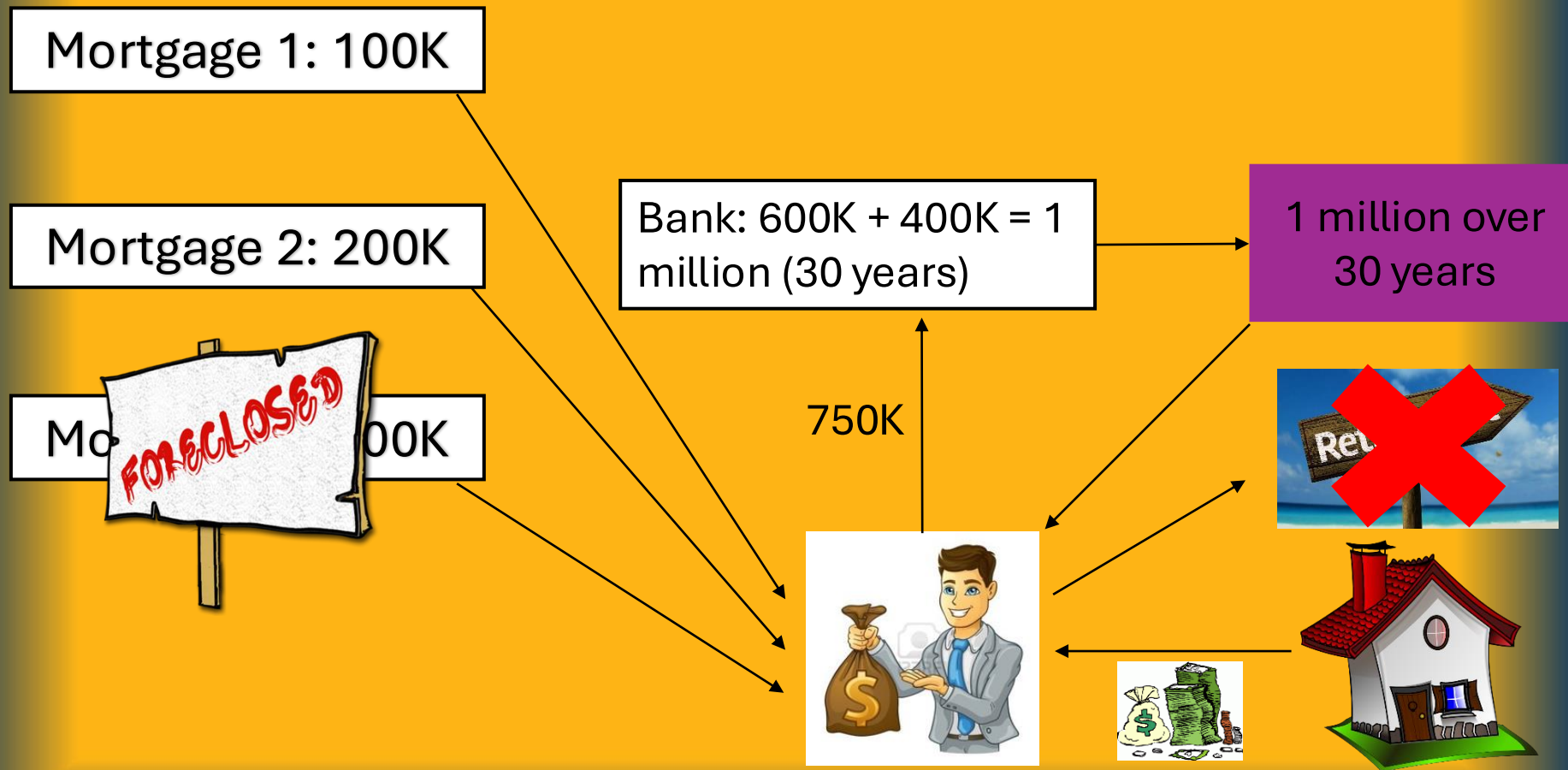
Housing Market Equilibrium

- Las Vegas Housing Prices Over Time

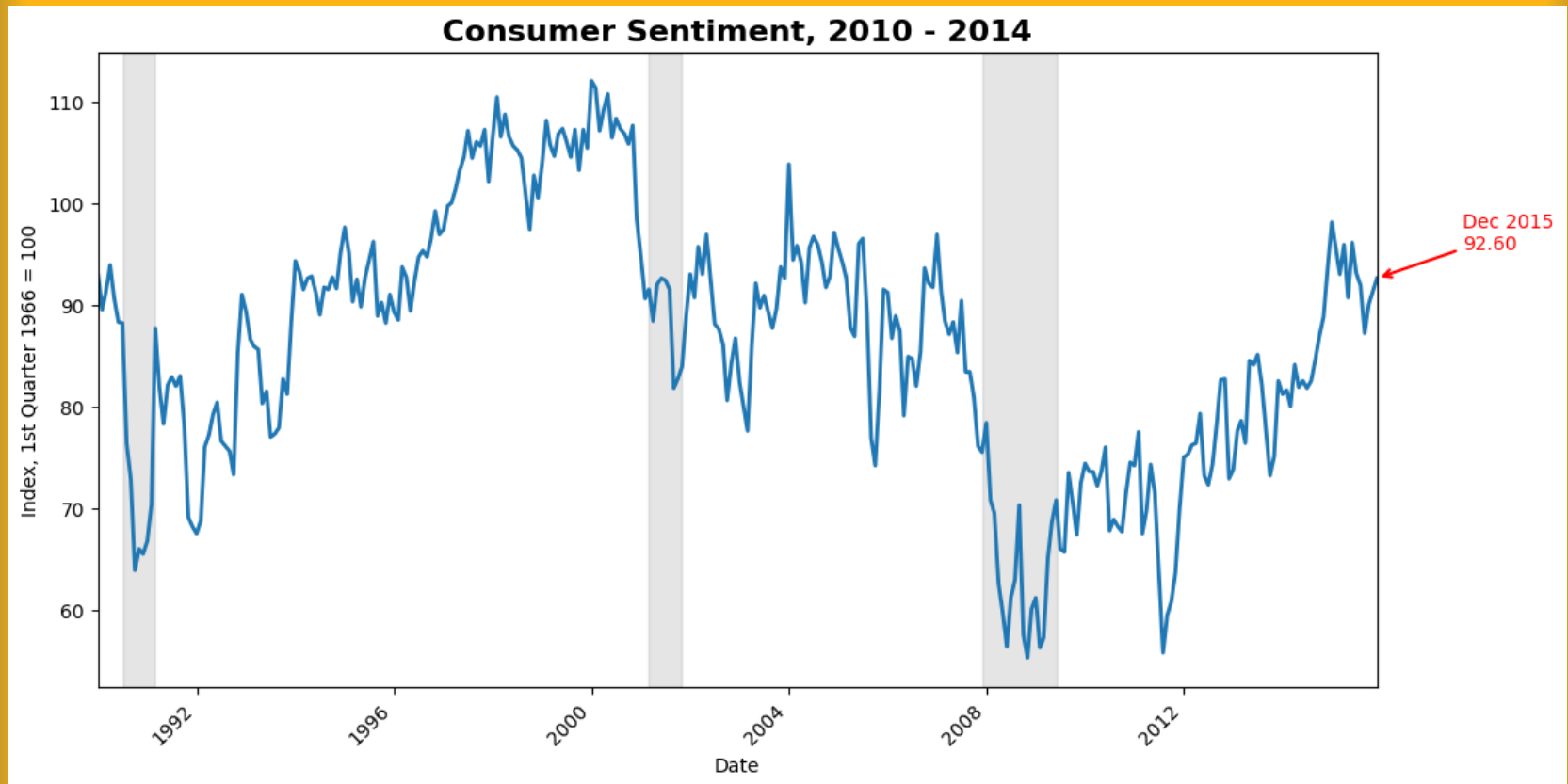


Consequences of the Housing Crash

- Mortgage Backed Securities

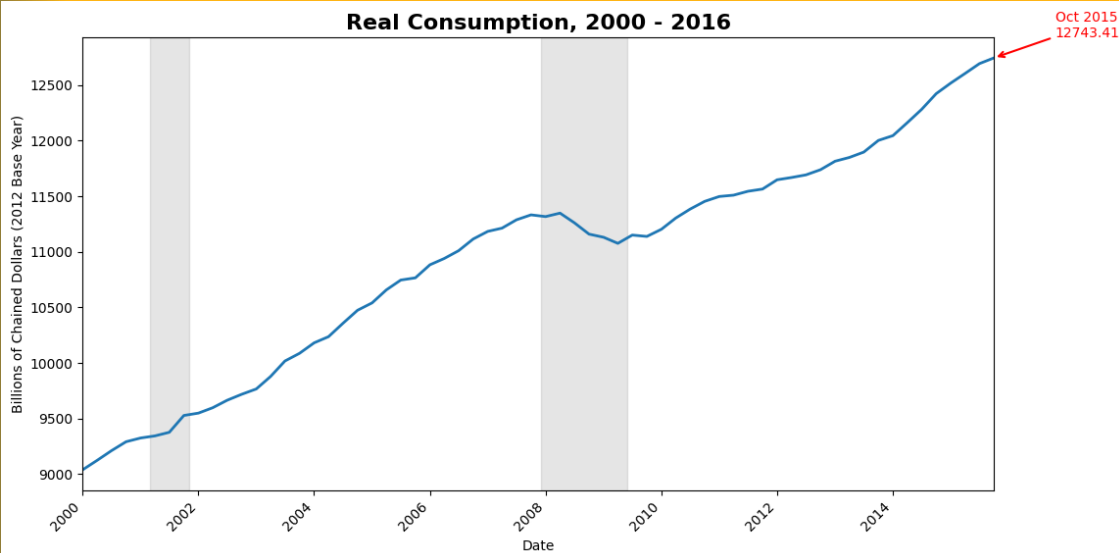


Consumer Confidence Decreased

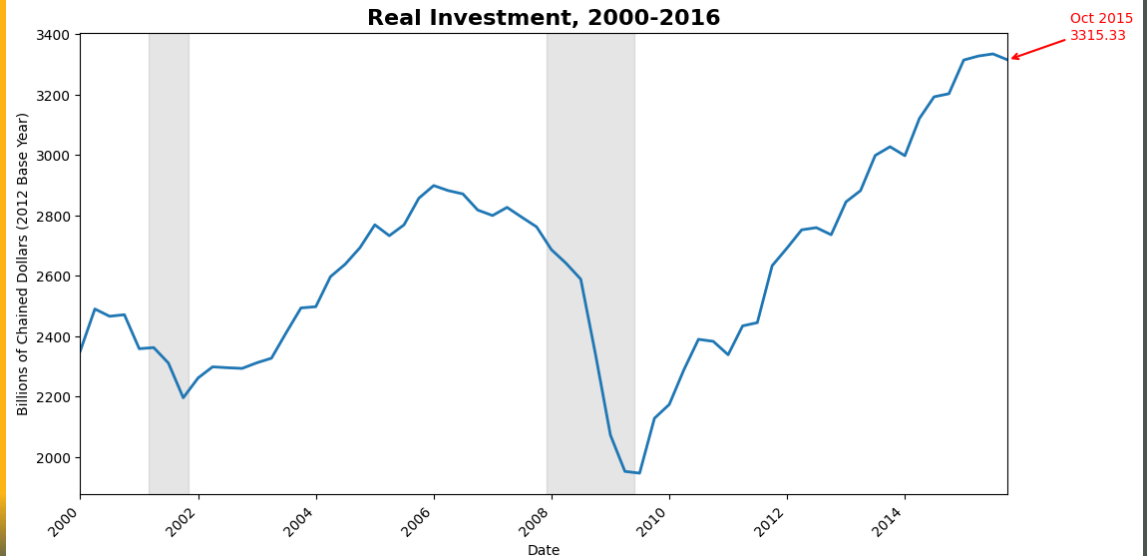


Housing: Decrease in AC and I

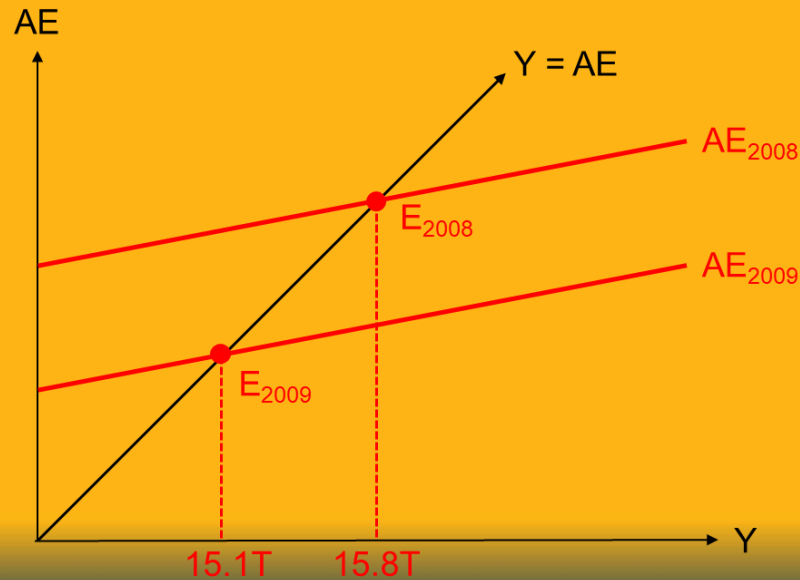
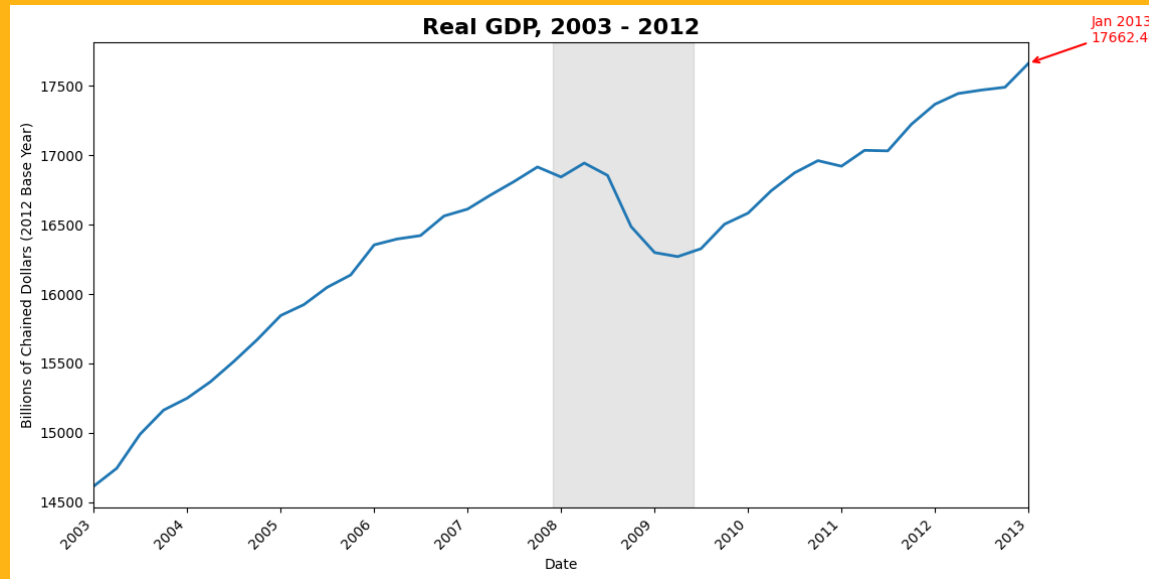
Real Consumption, 2000 - 2016



Real Investment, 2000-2016

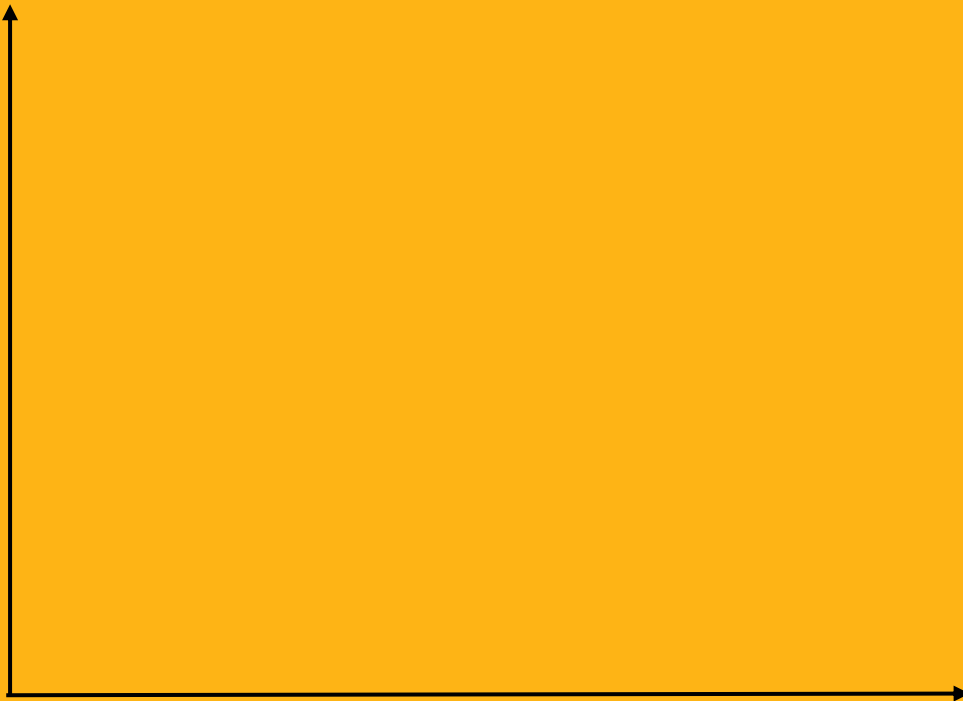


Great Recession



Changes in Equilibrium

- Start at $Y^* = 10,000$ (full-employment output)
Housing Crash! (assume $mpc = 0.8$)



Recovering from a Crash

- Option 1:

\$500 in machines = increase I^P by \$500 Incomes increase by \$500

 es (I^P)

Recovering from a Crash

- Option 1: Rich person buys \$500 worth of machines (I^P)

Round 1:
Y up \$500

\$500 in machines = increase I^P by \$500
Incomes increase by \$500

Round 2:
Y up \$400 more,
\$900 total

Spend $mpc \times 500 =$
 $0.80 \times 500 = \$400$
Consumption increase

Save $mps \times 500 =$
 $0.20 \times 500 = \$100$

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Incomes rise by \$400

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Incomes increase by \$500

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Y up \$400 more,
\$900 total

Spend $mpc \times 500 =$
 $0.80 \times 500 = \$400$
Consumption increase

Save $mps \times 500 =$
 $0.20 \times 500 = \$100$

Incomes rise by \$400

Round 3:
Y up \$320,
\$1,220 total

Spend $mpc \times 400 =$
 $0.80 \times 400 = \$320$
Consumption increase

Save $mps \times 400 =$
 $0.20 \times 400 = \$80$